Research Project Report: Sleep Apnea Detection from Single-Lead ECG

Aligning with UN Sustainable Development Goal 3 — Good Health and Well-being

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1. Executive Summary

Sleep apnea is a serious sleep disorder affecting an estimated **936 million people worldwide** (Benjafield et al., 2019). Traditional diagnosis requires expensive polysomnography (PSG) in sleep labs, creating a massive accessibility gap — especially in low-resource settings. This project proposes a **cost-effective, automated sleep apnea detection system using single-lead ECG signals**, which can be acquired from low-cost wearable devices. By leveraging signal processing and machine learning in MATLAB, this work directly contributes to **SDG 3.4** (reducing premature mortality from non-communicable diseases) and **SDG 3.8** (achieving universal health coverage).

Project Type: Comparative Machine Learning Study
Platform: MATLAB R2023b / R2024a
Estimated Duration: 4 weeks
Output: Research Paper + MATLAB Code + Results Dashboard

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2. SDG Alignment

SDG Target	Alignment	How This Project Contributes
3.4 —Reduce NCD mortality	Direct	Sleep apnea is linked to hypertension, heart disease, stroke, and diabetes. Early detection prevents downstream cardiovascular events.
3.8 —Universal health coverage	Direct	Single-lead ECG is orders of magnitude cheaper than full PSG. Enables screening in primary care and remote/rural settings.
3.b —R&D for diseases	Indirect	Contributes to open-source biomedical signal processing methods that can be replicated globally.
9.5 —Scientific research	Direct	Produces a reproducible, open methodology using public datasets.
10.2 —Social inclusion	Indirect	Wearable-based screening democratizes access to sleep disorder diagnosis.

SDG Indicator Tracked: 3.4.1 — Mortality rate attributed to cardiovascular disease, cancer, diabetes, or chronic respiratory disease.

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3. Datasets

3.1 Primary Dataset: PhysioNet Apnea-ECG Database

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Attribute	Detail
Source	PhysioNet (Goldberger et al., 2000)
Records	70 full overnight ECG recordings (8 recordings used for the 2000 PhysioNet/ CinC Challenge)
Sampling Rate	100 Hz
Duration	~8–10 hours per recording
Annotations	Minute-by-minute apnea labels (A = apnea, N = normal) + respiratory event annotations
Train/Test Split	Pre-defined: 35 training + 35 test recordings
Segment Count	~34,347 minute-level labeled segments (17,045 training + 17,302 test)
Access	Free, open-access via PhysioNet
Citation	PhysioNet/CinC Challenge 2000 — "Detection of Sleep Apnea from ECG"

Why this dataset?

- It is the **most cited benchmark** for apnea detection from ECG.
- Pre-defined train/test split ensures fair comparison with published literature.
- Single-lead ECG mimics real-world wearable device output.

Download Link: https://physionet.org/content/apnea-ecg/1.0.0/

3.2 Secondary Dataset: MIT-BIH Polysomnographic Database

Attribute	Detail
Source	PhysioNet
Records	18 full overnight polysomnographic recordings
Duration	~80+ hours total
Channels	4–7 channels including ECG (2 leads), EEG, respiration, oxygen saturation
Sampling Rate	250 Hz (ECG)
Annotations	Beat-by-beat ECG annotations + apnea/ hypopnea events
Use Case	Cross-validation and feature validation

Download Link: https://physionet.org/content/slpdb/1.0.0/

3.3 Optional Validation Dataset: UCD Sleep Apnea Database

Attribute	Detail
Source	University College Dublin
Records	25 full overnight polysomnograms
ECG Sampling	128 Hz, 3-channel Holter ECG
Annotations	Apnea/hypopnea events (AASM criteria)
Use Case	External validation to prove generalizability

Download Link: https://physionet.org/content/ucddb/1.0.0/

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4. Problem Statement

Research Question:
Can single-lead ECG signals be used to accurately detect sleep apnea on a minute-by-minute basis, and which feature extraction method (time-domain HRV vs. frequency-domain HRV vs. wavelet-based features) provides the best classification performance?

Hypothesis:
A hybrid feature set combining time-domain HRV and wavelet-based features will outperform individual feature categories for apnea detection from single-lead ECG.

Objective:

1. Preprocess raw ECG signals to remove noise and detect R-peaks.
2. Extract three categories of features: time-domain HRV, frequency-domain HRV, and wavelet entropy.
3. Compare classification performance of SVM, Random Forest, and Neural Network classifiers.
4. Evaluate using standard metrics: Accuracy, Sensitivity, Specificity, and AUC.
5. Validate findings on a secondary dataset (MIT-BIH PSG or UCDDB).

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5. Literature Review (Key Papers)

Year	Authors	Contribution	Dataset Used
2000	de Chazal et al.	Baseline method for PhysioNet Challenge; achieved 82.2% accuracy using HRV + EDR features	PhysioNet Apnea-ECG
2014	Mendez et al.	Review of ECG-derived respiration (EDR) techniques for apnea detection	Multiple PhysioNet
2017	Sharma et al.	Wavelet-based feature extraction + SVM; achieved 87.5% accuracy	PhysioNet Apnea-ECG
2019	Biswas et al.	Deep learning (1D-CNN) on raw ECG; achieved 91.2% accuracy	PhysioNet Apnea-ECG

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Table 5 – continued

Year	Authors	Contribution	Dataset Used
2020	Khandoker et al.	Review of HRV-based sleep apnea detection; emphasized clinical significance	Multiple
2022	Zhang et al.	Multi-scale entropy + LSTM; achieved 93.1% accuracy	PhysioNet Apnea-ECG
2024	Various	Recent trend toward transformer-based architectures and multi-modal fusion	PhysioNet + Private

Research Gap: Most recent papers use deep learning (CNN/LSTM) which requires large data and computational resources. There is limited work comparing **classical machine learning with engineered features** in a systematic, reproducible framework — especially for resource-constrained settings where interpretability matters.

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6. Methodology

6.1 System Architecture

```
Raw ECG Signal (100 Hz)
    ↓
[Preprocessing]
    • Bandpass Filter (0.5–45 Hz)
    • Baseline Wander Removal (Median Filter / Wavelet)
    • R-Peak Detection (Pan-Tompkins Algorithm)
    ↓
[Feature Extraction]
    ├── Time-Domain HRV Features (5 features)
    ├── Frequency-Domain HRV Features (3 features)
    └── Wavelet-Based Features (4 features)
    ↓
[Classification]
    ├── Support Vector Machine (SVM)
    ├── Random Forest (RF)
    └── Feedforward Neural Network (NN)
    ↓
[Evaluation]
    Accuracy, Sensitivity, Specificity, AUC, Confusion Matrix
```

6.2 Preprocessing (Week 1)

Step 1: Bandpass Filtering

- Remove high-frequency noise (muscle artifacts) and low-frequency baseline wander.
- MATLAB: `designfilt('bandpassiir', 'FilterOrder', 4, 'HalfPowerFrequency1', 0.5, 'HalfPowerFrequency2', 45, 'SampleRate', 100)`

- Apply: `filtfilt(b, a, ecg_signal)` for zero-phase filtering.

Step 2: Baseline Wander Removal

- Method A: Median filter (200 ms + 600 ms windows) — standard in ECG processing.
- Method B: Discrete Wavelet Transform (DWT) decomposition, remove approximation coefficients.
- MATLAB: `wdenoise(signal, 8, 'Wavelet', 'db6')`

Step 3: R-Peak Detection

- **Primary:** Pan-Tompkins algorithm (implemented via `findpeaks` or `ecgpeak` in Signal Processing Toolbox).
- **Validation:** Cross-check with PhysioNet-provided `.qrs` annotation files.
- MATLAB: `[pks, locs] = findpeaks(filtered_signal, 'MinPeakHeight', threshold, 'MinPeakDistance', 0.6*fs)`

Step 4: Beat Annotation & RR Interval Calculation

- Compute RR intervals from detected R-peaks.
- Remove ectopic beats and artifacts using adaptive thresholding.
- MATLAB: `diff(locs) / fs` to get RR intervals in seconds.

6.3 Feature Extraction (Week 2)

All features are computed on a **per-minute basis** (matching PhysioNet annotations).

A. Time-Domain HRV Features (5 features)

Feature	Description	MATLAB Implementation
Mean RR (mRR)	Average RR interval	`mean(rr_intervals)`
SDNN	Standard deviation of NN intervals	`std(rr_intervals)`
RMSSD	Root mean square of successive differences	`sqrt(mean(diff(rr_intervals).^2))`
pNN50	% of successive RR intervals differing by >50 ms	`sum(abs(diff(rr_intervals)) > 0.05) / length(rr_intervals)`
HR Mean	Mean heart rate	`60 ./ mean(rr_intervals)`

B. Frequency-Domain HRV Features (3 features)

Feature	Band	Description	MATLAB Implementation
LF Power	0.04–0.15 Hz	Low frequency (sympathetic + parasympathetic)	`pwelch(rr_interp, [], [], [0.04:0.001:0.15], 4)`
HF Power	0.15–0.4 Hz	High frequency (parasympathetic)	`pwelch(rr_interp, [], [], [0.15:0.001:0.4], 4)`
LF/HF Ratio	—	Sympathovagal balance	`LF / HF`

Note: RR intervals must be resampled to 4 Hz using cubic spline interpolation before spectral analysis.

- MATLAB: `interp1(rr_times, rr_intervals, new_times, 'cubic')`

C. Wavelet-Based Features (4 features)

Feature	Description	MATLAB Implementation
Wavelet Entropy (WE)	Shannon entropy of wavelet coefficient energies	Custom function using `wentropy`
Approximation Energy (A5)	Energy of 5th-level approximation coefficients	`sum(cA5.^2)`
Detail Energy Ratio (D3/D4)	Ratio of detail coefficients at scales 3 & 4	`sum(cD3.^2) / sum(cD4.^2)`
Wavelet Variance (WV)	Variance of wavelet coefficients at optimal scale	`var(cD4)`

- MATLAB: `[c, l] = wavedec(ecg_minute, 5, 'db4')` then extract `cA5`, `cD1`–`cD5`.

D. ECG-Derived Respiration (EDR) — Optional

Feature	Description	MATLAB Implementation
EDR Amplitude	R-peak amplitude modulation as proxy for respiration	Extract R-peak amplitudes, interpolate, compute peak-to-peak
EDR Rate	Breathing rate from EDR signal	`findpeaks` on interpolated EDR signal

6.4 Classification (Week 3)

Model 1: Support Vector Machine (SVM)

- **Kernel:** RBF (Radial Basis Function)
- **Hyperparameters:** BoxConstraint (C), KernelScale (gamma)
- **Tuning:** Bayesian Optimization or Grid Search
- MATLAB: `fitcsvm(X_train, y_train, 'KernelFunction', 'rbf', 'OptimizeHyperparameters', 'auto', 'HyperparameterOptimizationOptions', struct('Optimizer', 'bayesopt'))`

Model 2: Random Forest

- **Trees:** 100 (default, tune via OOB error)
- **Features per split:** sqrt(total features)
- MATLAB: `TreeBagger(100, X_train, y_train, 'Method', 'classification', 'OOBPrediction', 'on')`

Model 3: Feedforward Neural Network

- **Architecture:** [64, 32, 16] hidden layers with ReLU activation
- **Training:** Adam optimizer, 100 epochs, early stopping
- MATLAB: `fitcnet(X_train, y_train, 'LayerSizes', [64 32 16], 'Activations', 'relu', 'MaxEpochs', 100, 'ValidationData', {X_val, y_val})`

Feature Selection (Optional Enhancement)

- **Method:** ReliefF algorithm or Recursive Feature Elimination (RFE)
- MATLAB: `relieff(X, y, 10)` — ranks features by importance.

6.5 Evaluation Metrics

Metric	Formula / Description	Target
Accuracy	(TP + TN) / (TP + TN + FP + FN)	$> 85\%$
Sensitivity (Recall)	TP / (TP + FN)	$> 85\%$ (critical for medical screening)
Specificity	TN / (TN + FP)	$> 85\%$
Precision	TP / (TP + FP)	$> 80\%$
F1-Score	$2 \times$ (Precision $\times$ Recall) / (Precision + Recall)	$> 82\%$
AUC-ROC	Area under ROC curve	> 0.90
Kappa Statistic	Agreement beyond chance	> 0.70

Statistical Testing:

- McNemar's test to compare classifier performance (MATLAB: `mcmemar` or custom implementation).
- 10-fold cross-validation for robustness.

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7. MATLAB Toolboxes Required

Toolbox	Purpose	Functions Used
Signal Processing Toolbox	Filtering, peak detection, spectral analysis	`designfilt`, `filtfilt`, `findpeaks`, `pwelch`, `periodogram`
Wavelet Toolbox	Wavelet decomposition, denoising	`wavedec`, `wenergy`, `wentropy`, `wdenoise`
Statistics and Machine Learning Toolbox	Classification, feature selection, evaluation	`fitcsvm`, `TreeBagger`, `fitcnet`, `relieff`, `confusionmat`, `perfcurve`
Curve Fitting Toolbox	RR interval interpolation	`interp1` (built-in, but spline functions here)
Deep Learning Toolbox	(Optional) Advanced neural networks	`trainingOptions`, `trainNetwork`

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8. Expected Results (Based on Literature)

Method	Expected Accuracy	Expected AUC	Expected Sensitivity
Time-domain HRV only	78–82%	0.82–0.86	75–80%
Frequency-domain HRV only	76–80%	0.80–0.84	73–78%

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Table 12 – continued

Method	Expected Accuracy	Expected AUC	Expected Sensitivity
Wavelet features only	80–85%	0.85–0.89	78–83%
Hybrid (All features)	**85–90%**	**0.88–0.93**	**82–87%**
Hybrid + Feature Selection	86–91%	0.89–0.94	83–88%

Your Contribution: A systematic comparison showing which feature category contributes most, plus a validated hybrid model that is interpretable and deployable on low-cost hardware.

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9. Research Paper Structure

Title Suggestion

> *"Comparative Analysis of HRV and Wavelet-Based Features for Sleep Apnea Detection from Single-Lead ECG: A Machine Learning Approach"*

Abstract (~250 words)

- Background: Sleep apnea prevalence and diagnostic gap.
- Objective: Develop and compare ML models using ECG-derived features.
- Methods: PhysioNet Apnea-ECG dataset, three feature categories, three classifiers.
- Results: Expected accuracy, sensitivity, AUC.
- Conclusion: Implications for wearable-based screening and SDG 3.

1. Introduction (~1,000 words)

- Sleep apnea: definition, prevalence, health consequences.
- Current diagnostic limitations (PSG cost, accessibility).
- ECG as an alternative: physiological rationale (autonomic dysfunction during apnea).
- SDG 3 alignment: reducing health inequities.
- Research gap and objectives.

2. Related Work (~800 words)

- Review 5–7 key papers (see Literature Review table).
- Identify the gap: lack of systematic feature comparison with interpretable ML.

3. Materials and Methods (~1,500 words)

- Dataset description (PhysioNet Apnea-ECG, MIT-BIH PSG).
- Preprocessing pipeline with parameter justification.
- Feature extraction: mathematical definitions.
- Classification algorithms and hyperparameters.
- Evaluation protocol (train/test split, cross-validation, metrics).

4. Results (~1,000 words)

- Table: Feature importance ranking.
- Table: Classifier performance comparison.
- Figure: ROC curves for all models.
- Figure: Confusion matrices.
- Figure: Feature correlation heatmap.
- Statistical significance test results.

5. Discussion (~800 words)

- Interpretation of results.
- Why hybrid features outperform single categories.
- Clinical implications: wearable deployment.
- Limitations: dataset size, single-channel ECG, apnea vs. hypopnea distinction.
- SDG 3 impact: democratizing sleep disorder screening.

6. Conclusion (~300 words)

- Summary of findings.
- Contribution to SDG 3.
- Future work: real-time implementation, multi-night analysis, edge deployment.

References

- 20–30 citations (mix of foundational + recent 2020–2026 papers).

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10. 4-Week Execution Timeline

Week 1: Data Acquisition & Preprocessing

Day	Task	Deliverable
1	Download PhysioNet Apnea-ECG dataset; install WFDB toolbox	Dataset loaded in MATLAB
2	Explore data structure: `.dat`, `.hea`, `.apn` files	Data understanding document
3	Implement bandpass filter (0.5–45 Hz)	Filtered ECG signals
4	Implement baseline wander removal (median filter + wavelet)	Clean ECG signals
5	Implement Pan-Tompkins R-peak detection	R-peak locations for all recordings
6	Validate R-peaks against PhysioNet `.qrs` annotations; compute RR intervals	RR interval time-series

Table 13 – continued

Day	Task	Deliverable
7	**Buffer / Debug Day** – fix preprocessing issues	Clean, validated data ready

Week 2: Feature Extraction

Day	Task	Deliverable
8	Extract time-domain HRV features (mRR, SDNN, RMSSD, pNN50, HR Mean)	Feature matrix: Time-domain
9	Resample RR intervals to 4 Hz; compute frequency-domain features (LF, HF, LF/HF)	Feature matrix: Frequency-domain
10	Implement wavelet decomposition (db4, level 5); extract wavelet entropy & energies	Feature matrix: Wavelet
11	Extract EDR features (optional but recommended)	Feature matrix: EDR
12	Combine all features; normalize/standardize (z-score)	Final feature matrix (N × 12)
13	Compute feature correlation matrix; remove highly correlated features ($r > 0.9$)	Reduced feature set
14	**Buffer / Visualization Day** –plot feature distributions by class	Feature analysis plots

Week 3: Classification & Evaluation

Day	Task	Deliverable
15	Train SVM (RBF kernel) with Bayesian hyperparameter optimization	SVM model + performance metrics
16	Train Random Forest (100 trees) with OOB error monitoring	RF model + performance metrics

Table 15 – continued

Day	Task	Deliverable
17	Train Feedforward Neural Network (3 hidden layers)	NN model + performance metrics
18	Run 10-fold cross-validation for all models	Cross-validated metrics
19	Generate ROC curves, confusion matrices, precision-recall curves	Evaluation figures
20	Perform feature importance analysis (ReliefF + RF Gini importance)	Feature ranking table
21	**Buffer / Statistical Testing** –McNemar' s test, confidence intervals	Statistical comparison report

Week 4: Validation, Visualization & Paper Writing

Day	Task	Deliverable
22	(Optional) Validate best model on MIT-BIH PSG or UCDDB dataset	External validation results
23	Create all publication-quality figures (ROC, confusion matrices, feature heatmaps)	Figure folder (300 DPI PNG/EPS)
24	Write Introduction + Related Work sections	Draft: Sections 1–2
25	Write Methods section with all equations and parameters	Draft: Section 3
26	Write Results section with tables and figures	Draft: Section 4
27	Write Discussion + Conclusion + Abstract	Draft: Sections 5–6 + Abstract
28	Final editing, reference formatting (IEEE or APA), code documentation	Final paper + clean code

11. Key MATLAB Code Snippets

11.1 Load PhysioNet Data (WFDB Toolbox)

```
% Install WFDB Toolbox for MATLAB from PhysioNet
% URL: https://physionet.org/content/wfdb-matlab/

[signal, fs, tm] = rdsamp('a01', 'PhysioNetApneaECG');
ecg = signal(:, 1); % Channel 1 is ECG

% Load annotations
[ann, type, subtype, chan, num, comments] = rdann('a01', 'apn', 'PhysioNetApneaECG');
```

11.2 Pan-Tompkins R-Peak Detection

```
function [peaks, locs] = panTompkins(ecg, fs)
    % Bandpass filter
    [b, a] = butter(3, [5 15]/(fs/2), 'bandpass');
    filtered = filtfilt(b, a, ecg);

    % Differentiation
    diff_ecg = diff(filtered);

    % Squaring
    squared = diff_ecg .^ 2;

    % Moving window integration
    window = round(0.15 * fs);
    integrated = movmean(squared, window);

    % Peak detection
    minPeakDistance = round(0.6 * fs); % 600 ms
    minPeakHeight = 0.5 * max(integrated);
    [peaks, locs] = findpeaks(integrated, 'MinPeakHeight', minPeakHeight, ...
                              'MinPeakDistance', minPeakDistance);
end
```

11.3 Wavelet Feature Extraction

```
function features = extractWaveletFeatures(ecg_segment, fs)
    % DWT with db4, level 5
    [c, l] = wavedec(ecg_segment, 5, 'db4');

    % Extract coefficients
    cA5 = appcoef(c, l, 'db4', 5);
    cD5 = detcoef(c, l, 5);
    cD4 = detcoef(c, l, 4);
    cD3 = detcoef(c, l, 3);

    % Wavelet entropy
    energies = [sum(cA5.^2), sum(cD5.^2), sum(cD4.^2), sum(cD3.^2), ...
                sum(detcoef(c,l,2).^2), sum(detcoef(c,l,1).^2)];
```

```
    prob = energies / sum(energies);
    wavelet_entropy = -sum(prob .* log2(prob + eps));

    % Feature vector
    features = [sum(cA5.^2), sum(cD4.^2)/sum(cD3.^2), ...
                var(cD4), wavelet_entropy];
end
```

11.4 SVM Training with Optimization

```
% Prepare data
X = feature_matrix; % N x 12
y = labels; % N x 1 (categorical: 'A' or 'N')

% Train SVM with Bayesian optimization
svm_model = fitcsvm(X, y, 'KernelFunction', 'rbf', ...
    'OptimizeHyperparameters', 'auto', ...
    'HyperparameterOptimizationOptions', ...
    struct('Optimizer', 'bayesopt', 'MaxObjectiveEvaluations', 30));

% Cross-validation
cv_model = crossval(svm_model, 'KFold', 10);
loss = kfoldLoss(cv_model);
accuracy = 1 - loss;
```

11.5 ROC Curve & AUC

```
% Get prediction scores
[~, scores] = predict(svm_model, X_test);

% Compute ROC
[X_roc, Y_roc, T, AUC] = perfcurve(y_test, scores(:, 2), 'A');

% Plot
figure;
plot(X_roc, Y_roc, 'LineWidth', 2);
hold on;
plot([0 1], [0 1], 'k--');
xlabel('False Positive Rate');
ylabel('True Positive Rate');
title(['ROC Curve (AUC = ' num2str(AUC, '%.3f') ')']);
grid on;
```

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12. Potential Challenges & Mitigations

Challenge	Mitigation
R-peak detection errors	Validate against PhysioNet `.qrs` files; implement adaptive thresholding; manually inspect 5–10 segments.
Class imbalance (~50/50 in Apnea-ECG, but verify)	Use SMOTE if needed (`smote` in MATLAB); or use balanced class weights in SVM/RF.
Overfitting on small dataset	Strict train/test split; 10-fold CV; regularization in NN; limit model complexity.
Missing values in features	Impute with median; or remove minutes with < 3 valid RR intervals.
MATLAB license limitations	Use MATLAB Online (free for students) or Octave (limited toolbox support).
Paper rejection due to "incremental" work	Emphasize: (1) systematic feature comparison, (2) interpretability for clinical use, (3) SDG 3 framing, (4) reproducible open-source code.

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13. Target Venues for Publication

Venue Type	Examples	Timeline	Impact
IEEE Conference	IEEE EMBC, IEEE BIBE, IEEE ICASSP (biomedical track)	4–6 months	Good for early-stage work
Springer LNNS / LNEE	Biomedical Engineering conferences	3–5 months	Decent, peer-reviewed
MDPI Sensors / Bioengineering	Open access, fast review	2–4 months	Good visibility, APC fees
Elsevier Biomedical Signal Processing	Journal	6–12 months	Higher impact
ArXiv + GitHub	Preprint + code	Immediate	Visibility, no peer review

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14. SDG Impact Statement (For Paper)

> "Sleep apnea affects nearly 1 billion people globally, yet 80% remain undiagnosed due to the high cost and inaccessibility of polysomnography. This work demonstrates that single-lead ECG — acquirable from low-cost wearable devices — can detect sleep apnea with [XX]% accuracy using interpretable machine learning. By reducing diagnostic barriers, this approach aligns with **SDG 3.8 (Universal Health Coverage)** and **SDG 3.4 (Reducing Non-Communicable Disease Burden)**, enabling scalable screening in primary care and resource-limited settings."

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15. References

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Report prepared for: Sleep Apnea Detection from Single-Lead ECG Research Project
SDG Alignment: Goal 3 — Good Health and Well-being
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Timeline: 4 Weeks